

Why the clustering window uses the locally averaged weight

Validation note, branch thecov2 — October 2026

1. What is computed

Galaxies g at $\mathbf{x}_g$ with weights w_g , randoms r with weights w_r , $\alpha = \sum_g w_g / \sum_r w_r$:

$$F(\mathbf{x}) = \rho_w(\mathbf{x}) - \alpha \rho_{w,r}(\mathbf{x}), \quad \rho_w(\mathbf{x}) = \sum_g w_g \delta_D(\mathbf{x} - \mathbf{x}_g), \quad \hat{P} = (|\hat{F}|^2 - N_S) / \text{norm}. \quad (1)$$

The covariance needs window integrals $\int Wf$ and $\int Sf$ for smooth test functions f (e.g. $e^{-i\mathbf{q}\cdot\mathbf{x}}$), evaluated as sums over randoms.

2. Model of the catalogue (the one assumption)

Galaxies are a Poisson sample of $\bar{n}(\mathbf{x})[1 + \delta(\mathbf{x})]$. Each object carries a weight drawn from a local distribution $p(w|\mathbf{x})$, **independently of the other objects and of δ** (assumption A). Write $\langle w^p \rangle(\mathbf{x}) = \int w^p p(w|\mathbf{x}) dw$. Randoms are a Poisson sample of $n_r(\mathbf{x}) = \bar{n}(\mathbf{x})/\alpha$ whose weights are drawn from the same $p(w|\mathbf{x})$ (DESI randoms inherit the weights of random data objects).

3. Exact two-point function

Split the double sum $\sum_{g,g'}$ into distinct pairs and self-pairs. For distinct pairs A gives $\langle w_g w_{g'} \rangle = \langle w \rangle(\mathbf{x}) \langle w \rangle(\mathbf{y})$; the self-pairs carry w_g^2 :

$$\langle \rho_w(\mathbf{x}) \rho_w(\mathbf{y}) \rangle = m(\mathbf{x}) m(\mathbf{y}) [1 + \xi(\mathbf{x} - \mathbf{y})] + \delta_D(\mathbf{x} - \mathbf{y}) \bar{n} \langle w^2 \rangle(\mathbf{x}), \quad m(\mathbf{x}) \equiv \bar{n}(\mathbf{x}) \langle w \rangle(\mathbf{x}). \quad (2)$$

Adding the randoms, $\langle F(\mathbf{x}) F(\mathbf{y}) \rangle = m(\mathbf{x}) m(\mathbf{y}) \xi(\mathbf{x} - \mathbf{y}) + \delta_D(\mathbf{x} - \mathbf{y}) S(\mathbf{x})$ with $S = (1 + \alpha) \bar{n} \langle w^2 \rangle$. So **clustering** terms see the mean weighted density m , and the window is $W = m^2 = \bar{n}^2 \langle w \rangle^2$; only **shot-noise** terms see $\langle w^2 \rangle$.

4. Evaluating the window with randoms

A sum over randoms with each random's own weight as sampling weight is unbiased for m :

$$E \left[\alpha \sum_r w_r f(\mathbf{x}_r) \right] = \alpha \int n_r \langle w \rangle f = \int m f. \quad (3)$$

For $\int Wf = \int m^2 f$ the second factor $m(\mathbf{x}_r)$ must be supplied at the random. **Naive choice**, the random's own weight, $m(\mathbf{x}_r) \rightarrow \bar{n}(\mathbf{x}_r) w_r$:

$$E \left[\alpha \sum_r w_r^2 \bar{n} f \right] = \int \bar{n}^2 \langle w^2 \rangle f = \int m^2 f (1 + \text{var}(w) / \langle w \rangle^2). \quad (4)$$

The product $w_r \cdot w_r$ is a **self-pair**: it puts the shot-noise moment into the clustering window (+7% for DESI weights).

5. The approximation: replace $\langle w \rangle(\mathbf{x}_r)$ by a neighbour average

Use the k nearest **other** randoms, $\hat{w}_r = k^{-1} \sum_{j \in \mathcal{N}_k(r), j \neq r} w_j$. By A, w_r and the w_j are independent, and the estimator is linear in $\hat{w}_r$:

$$E[w_r \hat{w}_r] = \langle w \rangle(\mathbf{x}_r) \cdot \frac{1}{k} \sum_j \langle w \rangle(\mathbf{x}_j) = \langle w \rangle^2(\mathbf{x}_r) [1 + O(\ell_k^2 \nabla^2 \langle w \rangle / \langle w \rangle)], \quad (5)$$

where ℓ_k is the neighbourhood size (the term linear in $\mathbf{x}_j - \mathbf{x}_r$ cancels for a symmetric neighbourhood). Two remarks:

- **No noise bias**: because Equation 5 is linear in $\hat{w}_r$, the scatter of $\hat{w}_r$ does not bias $\int Wf$ (a $\text{var}(w)/k$ term appears only for higher powers of m , e.g. $\int m^4$, suppressed by $1/k \approx 3\%$ relative to the naive $\text{var}(w)/\langle w \rangle^2$).
- **Smoothing error**: $O(\ell_k^2 \nabla^2 \langle w \rangle)$: negligible where $\langle w \rangle$ varies on scales $\gg \ell_k$ ($\ell_k \approx 15h^{-1}\text{Mpc}$ for $k = 32$ in 3D; an angular neighbourhood, $\approx 0.1^\circ$, resolves sharp tile-completeness boundaries).

6. Result

With ρ_r the unweighted random density, $\alpha \rho_r = \bar{n}$, so the smooth mean weighted density and the window sums are

$$m(\mathbf{x}_r) = \alpha \rho_r(\mathbf{x}_r) \hat{w}_r, \quad \int Wf \approx \alpha \sum_r w_r m(\mathbf{x}_r) f(\mathbf{x}_r), \quad \int Sf \approx (1 + \alpha) \alpha \sum_r w_r^2 f(\mathbf{x}_r). \quad (6)$$

This is **NW** in `thecov.Tracer`: the averaged weight enters once, the random's own weight once (sampling), and the shot noise keeps the own weight squared.

When it fails. The averaging is only an estimator of $\langle w \rangle(\mathbf{x})$; the physics is assumption A. If weights correlate with each other or with δ (fiber-assignment/completeness weights depend on local target density and close pairs), then $\langle w_g w_{g'} \rangle \neq \langle w \rangle(\mathbf{x}_g) \langle w \rangle(\mathbf{x}_{g'})$ at small separation. That is a two-point effect: no one-point $m(\mathbf{x})$, averaged or not, can absorb it. Check: the GRF test with the high-resolution window and the smooth $\langle w \rangle$ agree to 0.2%.